

SMART INDIA HACKATHON 2024



- **Problem Statement ID** - SIH1620
- **Problem Statement Title**- Queuing models in OPDs/ availability of beds/ admission of patients. A hospital based solution is ideal which can be integrated with city wide module.
- **Theme**- Med tech / Bio Tech /Health Tech
- **PS Category**- Software
- **Team ID**- 1778
- **Team Name (Registered on portal)**- NextGen Coders

Description:

Technological solution as per queuing models in OPDs/availability of beds/admission of patients would be one area. Study of dispensation of various types of medicines/consumables and Inventory management modules at hospital level are key areas requiring support. NIC has already developed some modules but their implementation in Delhi is yet to be started. A hospital based solution is ideal which can be integrated with city wide module may be required.

PROPOSED SOLUTION

- This includes **real-time tracking** of bed occupancy, forecasting future availability.
- Admin dashboard for hospital staff to manage the dispensation of various types of medicines.
- Separate login for each doctor to check their appointment on current date.
- Providing a user friendly app in which staff nurse can see the **availability of beds, medicine inventory management**
- Patients can **book their appointment** using website for specific doctor and can check the status for availability of doctor
- Make an appointment **even without smartphone**. By using normal keypad phone anyone can book appointment by **sending SMS**.
- Visualize the inventory data using Graph, Chart etc.



Visualize the
inventory data



Book an
appointment with
keypad mobile



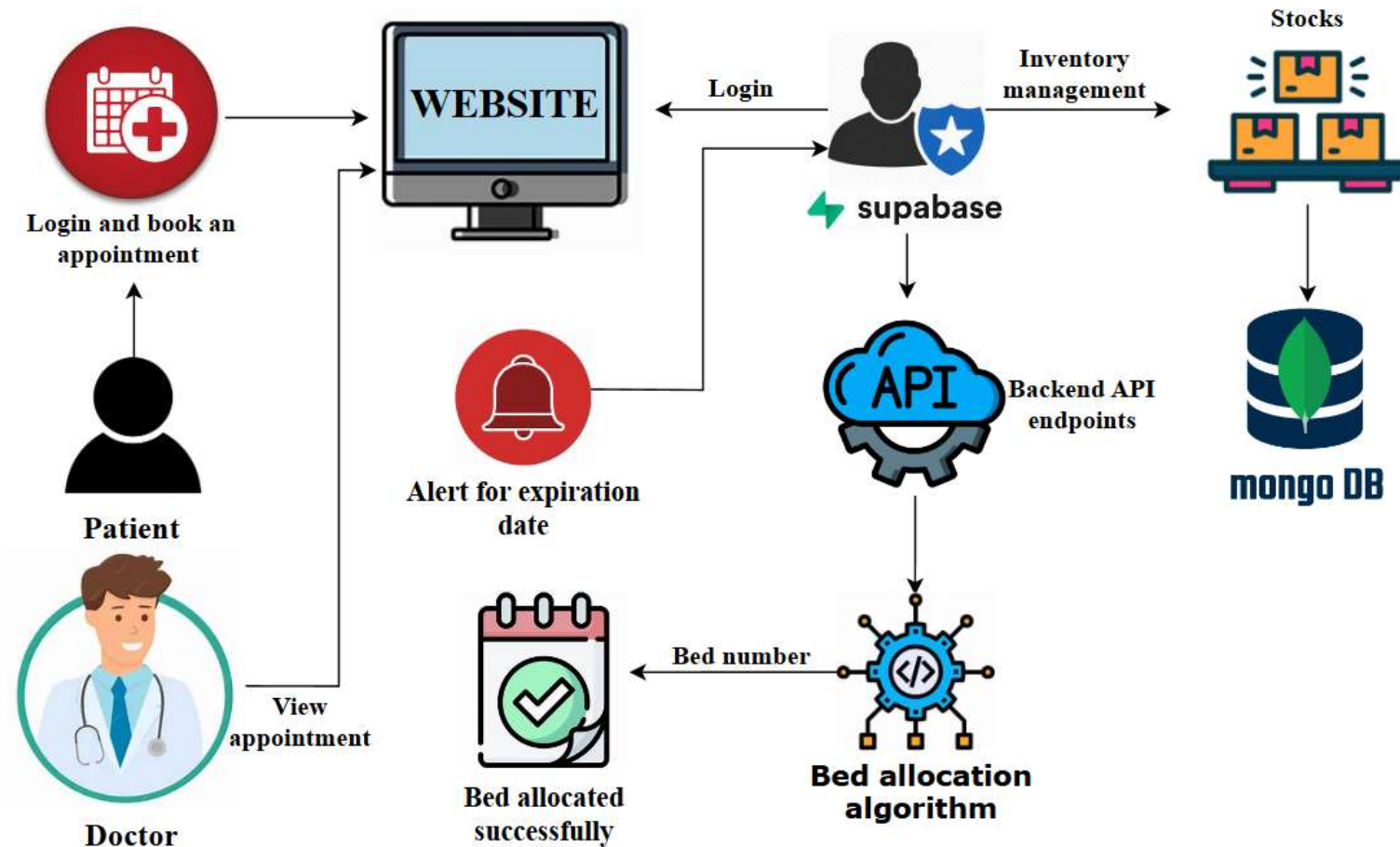
Four different login:

- Patient
- Head nurse
- Doctors
- Inventory admin

Comparison table:

FEATURES	HEALPROS	HOSPILOGIX	MOCDOC
APPOINTMENT & SCHEDULING	✓	✓	✓
BED ALLOCATION BY ALGORITHM	✓	✗	✗
INVENTORY MANAGEMENT SYSTEM	✓	✓	✗
ALERT FOR EXPIRATION DATE	✓	✗	✗
LOGIN FOR DOCTORS TO VIEW PATIENT DETAILS	✓	✗	✗
REMOTE MONITORING OF BED ALLOCATION	✓	✗	✗
BOOKING APPOINTMENT WITH KEYPAD PHONE	✓	✗	✗

Idea/Approach Chart



Analysis of the feasibility:

- Bed occupancy details on each floor can be easily obtained.
- Whole application is enclosed within the hospital only. No involvement of third person.
- Every hospital must have one computer at least for billing. So, using web application won't be a difficult one.

Strategies for overcoming these challenges:

- The login with **OTP (One-Time Password)** feature helps prevent unauthorized access.
- Assigning a bed number during patient admission prevents manual errors.

Potential challenges and risks:

- There are some risks, such as disclosing login credentials to other staff members working in the same hospital.
- Additionally, there is a possibility of miscommunication or selecting the wrong bed number when assigning beds manually.

TECHNOLOGICAL STACK:



tailwindcss



mongoDB.



supabase



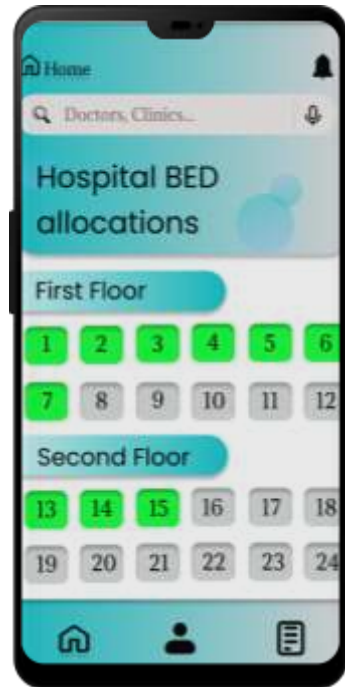
Potential impact on the target audience

- Significant Time Reduction
- Error Prevention
- Efficient Hospital Management
- Conflict-Free Bed Occupancy

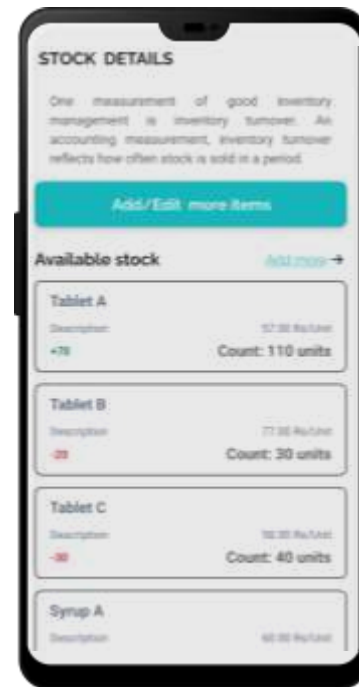
Benefits of the solution

- No patients can suffer by waiting.
- Reduced Administrative Burden
- Accessibility for All
- Lower Maintenance and Operational Costs

Bed occupancy status



Inventory stock details



Expiration alert



Appointment booking



Doctor's dashboard



Reference link:

<https://kgkhospital.com/book-an-appointment.html>

<https://www.v3labs.in/#team>

<https://www.medrot.com/blog/15-different-types-of-medical-beds-in-hospital/>

<https://www.nextgenesolutions.com/product/hospilogix>

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